MIL-G-36479a
26 October 1966
SUPERSEDING MIL-G-36479
dated 19 April 1966

MILITARY SPECIFICATION

GLOVES, X-RAY PROTECTIVE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of X-ray protective gloves, suitable for protecting hands and forearms of wearer from exposure to X-rays in military medical facilities.

1.2 Classification. Gloves covered by this specification shall be of the following types and sizes.

1.2.1 Types. Type I.
Type II.

1.2.2 Sizes. Large.
Medium.
Small.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6532

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

3.1 First Article. Unless otherwise specified, prior to the start of production of the contract quantity, two samples of the gloves shall be assembled, completely finished, and tested by the contractor. The samples shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the samples shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first articles, acquisition of materials, components, or commencement of production is at the sole risk of the contractor. Accepted samples shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the samples are not approved, the contractor shall submit a new sample for approval, or modify the existent samples so as to remove the deficiencies cited.

3.2 Material.

3.2.1 Type I. Gloves shall be manufactured from compounded milled neoprene.

3.2.2 Type II. Gloves shall be manufactured from lead impregnated vinyl plastic and shall have a soft absorbent cotton lining.

3.3 Construction. The gloves shall be supplied as a pair, one for right hand and one for left hand usage in the size specified. Each glove shall have five fingers.

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3.3.1 Type I. Type I gloves shall conform to the configuration as shown in figure 1. The gloves shall be constructed of several precisely fitting seamless plies, each of which shall be capable of sliding with respect to adjacent plies. Each seamless ply shall be built up by the multi-dip cement process. The glove plies shall be impregnated with lead, with the exception of the inner and outer glove surface plies which shall be unleaded. Each glove, including the cuff, shall be at least 13-1/4 inches long overall and shall be contour shaped to conform to the normal relaxed position of the hand. A bead rolled from the inner material shall be as shown in figure 1.

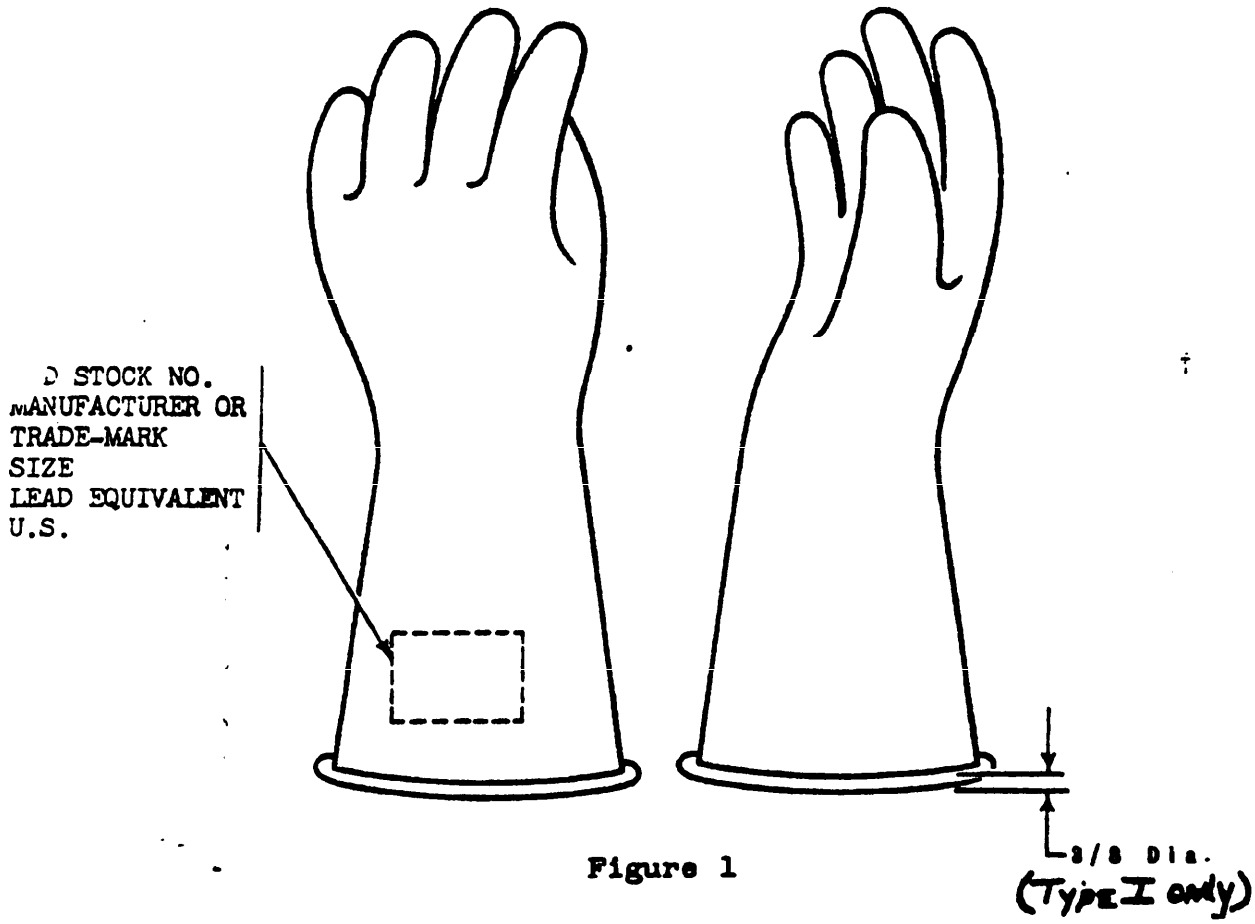


Figure 1

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3.3.2 Type II. Type II gloves shall be injection molded with no seams. Each glove including the cuff, shall be a minimum of 15 inches long overall.

3.3.3 Accessories.

3.3.3.1 Type I. Each pair of Type I gloves shall be furnished with two pairs of white cotton gloves to serve as absorbent inner liners.

3.3.3.2 Type II. Two pairs of green, slip-on external cotton sheaths with stretch cuffs shall be supplied with each pair of Type II gloves. The sheaths shall be made from washable, color fast, durable cotton and shall be held taut over the ends of the gloves by the stretch cuffs.

3.4 Performance. The design of the gloves shall be such as to provide sufficient dexterity and flexibility to the wearer to permit operation of X-ray apparatus controls and palpation of patients. The lead impregnation shall provide uniform protection to the wearer of not less than 0.5 millimeter (mm.) of lead when tested as specified in 4.4.1. Any density reading on the films denoting a lead equivalent below 0.5 mm. shall be cause for rejection except that any individual area smaller than 2 square mm. or a summation of all such areas amounting to not more than 12 square mm. may have a lead equivalency below 0.5 mm. but not less than 0.25 mm.

3.4.1 Type I. The internal and external surfaces of the gloves shall be smooth, shall be easily washable and shall be chemically resistant to soaps, antiseptecs, and desinfectants used in hospitals for rubber goods.

3.4.2 Type II. The gloves shall withstand continuous hard use and repeated flexing without cracking, breaking, or deterioration after testing as specified in 4.4.2.

3.5 Identification marking. Each glove shall be legibly and permanently marked on the back of the cuff, as shown in figure 1, with the following information:

Federal stock number.
Manufacturer's name or registered trademark.
Contract or purchase order number.
Size (small, medium, or large).
Lead equivalent in terms of mm.
U.S.

Markings secured by adhesives shall not be used.

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3.6 Service data. The contractor shall enclose, with each pair of gloves, two copies of an instruction sheet outlining the proper use and care of the gloves. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

3.7 Workmanship. Workmanship shall be first class throughout. Gloves shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (defects per hundred units)
Major defects	II	1.0
Total (major and minor defects combined)	II	4.0

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The gloves shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major		
	101	Gloves not sufficiently flexible to permit operation of X-ray controls.
	102	Gloves not free of cracks, cuts, tear, holes, perforation, or blisters.
	103	Gloves not a matched pair.
	104	Cotton glove inserts or sheaths not supplied.
	105	Size not as specified.
Minor		
	201	Identification marking missing, incomplete, or illegible.
	202	Service data not supplied.

*Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Lead equivalent test. One percent of the glove pairs in the lot (to the nearest whole number) shall be subjected to this test. Specimens prepared from each pair of gloves shall be radiographed alongside a suitable lead penetrometer, graduated in steps of 0.1 mm., on the front of a 14 by 17 inch cassette. Specimens shall be prepared from each finger, the palm and the back of each glove, and shall include the full thickness of material protecting the hand. Size of specimens shall be as large as is practicable, consistent with the area available on the front of the cassette, to obtain a radiographic density indication on the film for each specimen prepared from a single glove. The cassettes shall be exposed at 75 to 100 kilovolt-peak (KVP), using intensifying screens and suitable film to obtain a density of approximately 1.0 at the 0.5 mm. step of the lead penetrometer. The processed films shall be examined by means of a suitable densitometer.

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4.4.2 Temperature tests. Type II gloves shall be subjected to a high temperature test and a low temperature test. Tolerances on each temperature shall be ± 5°F. At the end of each test, the gloves shall be put on, repeatedly flexed, and examined for cracks, breaks, and deterioration.

4.4.2.1 Low temperature. The low temperature test shall be at -30°F. for 12 hours followed by an ambient of 72°F. for a minimum of 6 hours.

4.4.2.2 High temperature. The high temperature test shall be at +140°F. for 72 hours followed by an ambient of 72°F. for a minimum of 6 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One pair of type I gloves and two pairs of white cotton gloves, or one pair of type II gloves and two pairs of green cotton sheaths, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. The rubber gloves shall be supported by stuffing with glove spreader, paper, cotton, or cloth. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following.

(a) Number and date of this specification.

(b) Selection of applicable levels of packing (see 5.1 and 5.2).

(c) Where First Article shall be sent (see 3.1).

(d) Where service data shall be sent (see 3.6).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DH

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

CATEGORY II

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify):
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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